# US Health Insurance Exploratory Data Analysis

A Python-based investigation of medical insurance charges

## Project objective

Explore how medical insurance charges vary across age, BMI, smoking status, sex, region and number of children. The goal is to identify useful patterns in the dataset and explain them in plain language.

**Tools:** Python, Pandas, NumPy and Matplotlib

**Dataset:** US Health Insurance Dataset, Kaggle

**Dataset link:** https://www.kaggle.com/datasets/teertha/ushealthinsurancedataset

**Main takeaway: smoking status showed the largest difference in average insurance charges among the factors examined. Age and BMI also showed patterns, especially when studied alongside smoking status.**

This report summarizes the exploratory analysis shared in the project conversation. Findings describe associations in this dataset and should not be interpreted as proof of cause and effect.

# 1. Executive Summary

The dataset initially contained 1,338 records and seven columns. One exact duplicate row was removed, leaving 1,337 records. No missing values or negative values were found in the numerical columns checked. The analysis explored the distribution of each variable, compared charges across groups, examined age and BMI patterns, reviewed outliers, and calculated correlations.

| Area | Key result |
|---|---|
| Cleaned data | 1,337 rows; 7 original columns plus 2 derived grouping columns; no missing values or duplicate rows. |
| Typical charges | Mean: $13,279.12; median: $9,386.16. The mean being higher than the median indicates a right-skewed distribution. |
| Smoking status | Mean charges: $32,050.23 for smokers versus $8,440.66 for non-smokers. |
| Age | Mean age: 39.22; observed range: 18–64; correlation with charges: 0.298. |
| BMI | Mean BMI: 30.66; observed range: 15.96–53.13; overall correlation with charges: 0.198. |
| Children | Correlation with charges: 0.067, suggesting almost no linear relationship across the full dataset. |
| Regions | Southeast had the highest overall mean charges ($14,735.41) and the highest smoker share (25%). |
| Outliers | IQR method flagged 9 BMI observations (0.67%) and 139 charge observations (10.40%); these were retained. |

### Overall conclusion

Smoking status was the clearest group-level difference in charges. Charges generally rose with age, and BMI had a much stronger positive correlation with charges among smokers (0.806) than among non-smokers (0.084). Regional and sex differences were smaller or less consistent. The results are descriptive, not causal.

# 2. Dataset and Cleaning

The dataset includes age, sex, BMI, number of children, smoking status, region and medical insurance charges. Charges are the outcome we focused on because the project asks which characteristics are associated with higher or lower charges.

| Check | Result | Why it mattered |
|---|---|---|
| Dataset shape | 1,338 rows × 7 columns initially | Established the size and structure of the data. |
| Data types | 4 numerical columns and 3 categorical columns | Confirmed which summaries and plots were suitable. |
| Unique values | Age: 47; sex: 2; BMI: 548; children: 6; smoker: 2; region: 4; charges: 1,337 after cleaning | Helped understand whether columns behaved like numerical measures or categories. |
| Missing values | 0 in every original column | No missing-value treatment was needed. |
| Duplicate check | 1 exact duplicate row | The repeated record was removed to avoid counting the same row twice. |
| After cleaning | 1,337 rows × 7 original columns | Confirmed the duplicate was removed. |
| Negative values | 0 in age, BMI, children and charges | No negative numerical values were detected in the checked columns. |

The duplicate was found at original indices 195 and 581. Both rows contained the same values: age 19, male, BMI 30.59, zero children, non-smoker, northwest, charges $1,639.5631. The working dataset was called df_clean. Later, age_group and bmi_group were added for comparisons, giving a final shape of 1,337 × 9.

# 3. Understanding the Variables

The first plots and descriptive statistics showed what the dataset looks like before comparing groups. Histograms were used for numerical variables, while bar charts were used for categorical variables.

### Insurance charges

Mean charges were $13,279.12, while the median was $9,386.16. The mean is much higher than the median because a smaller group of people has very high charges. The histogram showed a long right tail, with most observations at lower charges and fewer observations reaching above $40,000. The maximum was $63,770.43 and the minimum was $1,121.87.

### Age

Age ranged from 18 to 64. The mean was 39.22 and the median was 39. The histogram showed individuals across the age range, with a noticeably large group around age 18. There were no age outliers under the IQR method.

### BMI

BMI ranged from 15.96 to 53.13. The mean was 30.66 and the median was 30.40. Most values clustered around the upper 20s and 30s, with a smaller number of high BMI values. Nine observations fell outside the IQR bounds.

### Number of children

The mean number of children was 1.1 and the median was 1. Counts were: 0 children, 573 people; 1, 324; 2, 240; 3, 157; 4, 25; and 5, 18. Most people had zero to two children, while groups with four or five children were small.

### Categorical variables

| Variable | Counts | What it tells us |
|---|---|---|
| Sex | Male: 676 initially, female: 662 initially. After duplicate removal: male 675, female 662. | The sample was nearly balanced by sex. |
| Smoker | No: 1,064 initially; yes: 274 initially. After cleaning: no 1,063; yes 274. | Most individuals were non-smokers, so group sizes were unequal. |
| Region | Southeast: 364 initially; southwest: 325; northwest: 325; northeast: 324. After cleaning: southeast 364, southwest 325, northwest 324, northeast 324. | Regional sample sizes were broadly similar. |

### Why these plots mattered

The distributions revealed the scale and shape of each variable, showed that charges were right-skewed, and highlighted that the smoker group was smaller than the non-smoker group. This context is important when comparing averages and interpreting later charts.

# 4. Charges by Individual Factors

### Smoking status

| Smoking status | Count | Mean charges | Median charges | Minimum | Maximum |
|---|---|---|---|---|---|
| Non-smoker | 1,063 | $8,440.66 | $7,345.73 | $1,121.87 | $36,910.61 |
| Smoker | 274 | $32,050.23 | $34,456.35 | $12,829.46 | $63,770.43 |

The box plot showed a clear separation: smokers generally had much higher charges than non-smokers. The average for smokers was about 3.8 times the non-smoker average. This was the largest and clearest group difference identified in the analysis. High charges among non-smokers also existed, so smoking status did not explain every case.

## Sex

After cleaning, the mean was $12,569.58 for females and $13,975.00 for males. The medians were very close: $9,412.96 for females and $9,377.90 for males. The box plots overlapped substantially, suggesting sex alone did not separate charge levels as clearly as smoking status.

## Region

| Region | Count | Mean charges | Median charges |
|---|---|---|---|
| Northeast | 324 | $13,406.38 | $10,057.65 |
| Northwest | 324 | $12,450.84 | $8,976.98 |
| Southeast | 364 | $14,735.41 | $9,294.13 |
| Southwest | 325 | $12,346.94 | $8,798.59 |

The Southeast had the highest mean charges, but the Northeast had the highest median. This difference suggests that very high charges in some observations can raise the Southeast mean. Region alone did not show a simple ranking that applies to every individual.

## Number of children

Mean charges were highest for people with three children ($15,355.32) and lowest for those with five ($8,786.04). However, the five-child group contained only 18 people, so its average is less stable. Charges did not rise consistently with the number of children. The overall correlation between children and charges was only 0.067.

# 5. Relationships and Combined Analysis

## Age versus charges

The scatter plot showed a general upward pattern: older individuals often had higher charges. The correlation was 0.298, a weak positive linear relationship. The points also formed visible bands, suggesting that another factor, particularly smoking status, was separating groups with different charge levels.

## BMI versus charges

The overall scatter plot had a wide spread, and the overall correlation was 0.198. By itself, BMI had only a weak positive linear relationship with charges. The plot separated more clearly after points were grouped by smoking status.

## Age and smoking status together

| Age group | Non-smoker mean | Smoker mean |
|---|---|---|
| 18–29 | $4,426.99 | $27,518.04 |
| 30–39 | $6,337.36 | $30,271.25 |
| 40–49 | $9,183.34 | $32,654.72 |
| 50–64 | $13,430.90 | $38,748.35 |

Average charges increased across age groups for both smokers and non-smokers. Smokers had much higher average charges in every age group. The gap was not limited to older people, although the highest smoker average was in the 50–64 group.

## BMI and smoking status together

| BMI group | Non-smoker mean | Smoker mean | Group size |
|---|---|---|---|
| Below 25 | $7,515.71 | $19,839.28 | 245 |
| 25 to 30 | $8,257.96 | $22,495.87 | 386 |
| Above 30 | $8,855.53 | $41,557.99 | 706 |

Among non-smokers, average charges increased gradually across the BMI groups. Among smokers, the above-30 BMI group had a much higher average ($41,557.99) than the other smoker groups. The BMI-charge correlation was 0.084 among non-smokers and 0.806 among smokers. This is a strong difference in the pattern between the two groups, but it does not prove that BMI or smoking caused the charges.

## Correlation matrix

| Variable | Correlation with charges | Plain-language reading |
|---|---|---|
| Age | 0.298 | Weak positive linear relationship; strongest among the numerical variables examined. |
| BMI | 0.198 | Weak positive linear relationship overall. |
| Children | 0.067 | Very weak linear relationship. |

Age and BMI had a correlation of 0.109 with each other; age and children, 0.042; BMI and children, 0.013. These values are close to zero. The matrix only included numerical variables, so it did not directly encode smoking status, sex or region.

# 6. Outlier Analysis

| Variable | IQR lower bound | IQR upper bound | Outlier count | Share of cleaned data |
|---|---|---|---|---|
| Age | -9.00 | 87.00 | 0 | 0.00% |
| BMI | 13.67 | 47.32 | 9 | 0.67% |
| Charges | -13,120.72 | 34,524.78 | 139 | 10.40% |

The IQR method flags values far from the middle half of the data. It found no age outliers, nine BMI outliers and 139 high-charge outliers. These observations were retained because an outlier is not automatically a mistake. In particular, high insurance charges may be real and are important to understanding the dataset. The negative lower bounds are mathematical cutoffs, not evidence of negative observations.

# 7. Regional Smoking Comparison

| Region | Non-smokers | Smokers | Smoker share |
|---|---|---|---|
| Northeast | 257 | 67 | 20.68% |
| Northwest | 266 | 58 | 17.90% |
| Southeast | 273 | 91 | 25.00% |
| Southwest | 267 | 58 | 17.85% |

The Southeast had the highest smoker share (25%) and the highest overall mean charges. This makes smoking composition a possible contributor to the regional difference, but this comparison alone cannot establish how much it explains.

### Within-region charge comparisons by smoking status

The range between the highest and lowest regional average was $1,146.25 among non-smokers and $5,171.46 among smokers. Regional average charges therefore varied more across regions within the smoker group. This is a descriptive comparison, not proof that region changes the effect of smoking.

# 8. Which Factors Were Most Associated with Charges?

| Factor | Strength of pattern in this analysis | Reason |
|---|---|---|
| Smoking status | Largest observed group difference | Mean charges were $32,050.23 for smokers and $8,440.66 for non-smokers. |
| Age | Weak positive overall relationship; clearer pattern within groups | Mean charges increased across age groups for both smokers and non-smokers. |
| BMI combined with smoking | Strong pattern among smokers | BMI-charge correlation was 0.806 among smokers versus 0.084 among non-smokers; smokers with BMI above 30 had a high average. |
| Region | Moderate descriptive differences | Southeast had the highest overall mean and smoker share, but regional patterns were not uniform. |
| Sex | Smaller difference | Male mean was higher, but male and female medians were almost the same. |
| Number of children | Little overall linear relationship | Correlation was 0.067; group averages did not rise consistently and some groups were small. |

**The phrase “most associated” is more accurate than “caused the most.” This EDA compares patterns in observed data; it does not control for every factor or establish causation.**

# 9. Practical Recommendations

1. Treat smoking status as a priority variable for further investigation because the average charge difference was large in this sample.

2. Examine age and BMI alongside smoking status instead of interpreting them only as separate variables. The combined charts revealed patterns that the overall correlations did not show.

3. Investigate regional differences within comparable groups before drawing conclusions from overall regional averages.

4. Keep plausible high-charge observations in the analysis unless there is evidence of a data error. Document the IQR flags rather than automatically deleting them.

5. Avoid strong conclusions from small groups, especially people with four or five children.

6. If this work is later extended, consider a statistical model that evaluates multiple factors together. That would be a separate step beyond this EDA-only project.

## 10. Limitations

• The dataset has 1,337 records after cleaning and may not represent every insured person or the wider US population.

• The analysis is observational and cannot prove that any factor caused higher charges.

• Correlation measures linear association and may miss more complex relationships or group differences.

• The smoker and non-smoker groups are unequal in size.

• Some age, BMI and children groups have relatively few observations.

• Charges are the recorded outcome in this dataset; the analysis does not establish how insurers actually set premiums or how medical need was assessed.

## 11. Final Conclusion

The EDA found that smoking status had the clearest association with medical insurance charges in this dataset. Charges also tended to increase with age. BMI had a weak relationship with charges overall but a much stronger relationship among smokers. Regional average charges differed, with the Southeast highest, while sex and number of children showed weaker or less consistent patterns. The data had no missing values after inspection, one exact duplicate was removed, and plausible outliers were retained. These findings provide a clear descriptive overview and a basis for further investigation, without claiming cause and effect.

Tools used: Pandas for inspection, cleaning, grouping and summaries; NumPy for numerical support; Matplotlib for histograms, bar charts, scatter plots, box plots and a correlation heatmap.